

Prague University of Economics and Business
Faculty of Informatics and Statistics



Predicting successful player alignments in volleyball using machine learning

BACHELOR THESIS

Study program: Data Analytics

Author: Igor Čurila

Supervisor: Ing. Ondřej Sokol, Ph.D.

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Abstrakt

Tato bakalářská práce se zabývá použitím technik strojového učení k analýze a předvídání účinnosti obranných formací ve volejbale. Na základě ručně anotovaného souboru dat 300 her z profesionálních volejbalových zápasů byly vyvinuty a vyhodnoceny tři různé modely strojového učení - logistická regrese, náhodný les a XGBoost. Z nich dosáhl nejvyšší přesnosti model XGBoost, ale všechny modely prokázaly solidní výkonnost, což naznačuje, že i s relativně malým a strukturovaným souborem dat lze provádět smysluplné předpovědi. Práce zdůrazňuje důležitost výběru příznaků a přípravy dat se zaměřením na proměnné dostupné před útokem, jako jsou pozice obránce, typ útoku, kvalita nahrávky a detaily bloku. Analýza důležitosti příznaků odhalila, že typ útoku, efektivita bloku a zejména pozice obránce 6 jsou rozhodujícími faktory ovlivňujícími úspěšnost obrany. Výsledky ukazují, že optimální obranné postavení není statické, ale mělo by se dynamicky přizpůsobovat v závislosti na konkrétní herní situaci, například na tom, zda se blok dotkne míče. Ačkoli byl soubor dat omezen co do velikosti a rozsahu, tento výzkum ukazuje, že strojové učení může trenérům a hráčům poskytnout cenné taktické poznatky. Díky větším souborům dat a automatizovaným metodám sběru dat mají tyto přístupy potenciál podpořit rozhodování v reálném čase a rozvoj strategie ve volejbale, což v konečném důsledku přispěje ke zlepšení výkonnosti týmu a hlubšímu pochopení hry.

Klíčová slova

volejbal, obranné formace, strojové učení, sportovní analytika

Abstract

This bachelor thesis investigates the application of machine learning techniques to analyze and predict the effectiveness of defensive formations in volleyball. Using a manually annotated dataset of 300 plays from professional volleyball matches, three different machine learning models—logistic regression, random forest, and XGBoost—were developed and evaluated. Among these, the XGBoost model achieved the highest accuracy, but all models demonstrated solid performance, suggesting that even with a relatively small and structured dataset, meaningful predictions can be made. The study emphasizes the importance of feature selection and data preparation, focusing on variables available before the attack, such as defender positions, attack type, set quality, and block details. Feature importance analysis revealed that the type of attack, block effectiveness, and especially the position of defender 6 are crucial factors influencing defensive success. The results show that optimal defensive positioning is not static but should adapt dynamically depending on the specific game situation, such as whether the block touches the ball. Although the dataset was limited in size and scope, this research demonstrates that machine learning can provide valuable tactical insights for coaches and players. With larger datasets and automated data collection methods, these approaches have the potential to support real-time decision-making and strategy development in volleyball, ultimately contributing to improved team performance and a deeper understanding of the game.

Keywords

volleyball, defensive formations, machine learning, sports analytics

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Introduction

As a long-time volleyball player and a student of data analytics, I have always been interested in how data and statistics can help us better understand and improve sports performance. During my years of playing volleyball, I noticed how much the positioning of players-especially in defensive situations-can influence the outcome of a match. This observation led me to wonder whether it would be possible to analyze these positions in a more objective way and to find patterns that could help coaches and players make better tactical decisions.

With the recent advances in artificial intelligence and machine learning, there are now more opportunities than ever to analyze sports data in new and meaningful ways. While sports like football and basketball already make extensive use of player tracking and computer vision for tactical analysis, volleyball is still a field with a lot of untapped potential for such methods. I chose this topic for my thesis because it brings together my two main interests: volleyball and data analysis. I believe that using machine learning to study player positioning can not only deepen my own understanding of the game, but also provide useful insights for the volleyball community.

The main goal of my thesis is to explore the use of machine learning for analyzing the positioning of defensive players in volleyball and to predict whether a given formation is successful based on game results. To do this, I created and annotated my own dataset, developed a model for classifying successful and unsuccessful defensive formations, and visualized the results. The thesis is divided into several sections: first, I review the necessary background about volleyball and related research; next, I describe how the dataset was collected and processed; then, I explain the design and implementation of the machine learning model; this is followed by a presentation and discussion of the results; and finally, I offer some practical recommendations and summarize the main conclusions.

1. Literature Review

Machine learning and data analysis have started to appear more and more often in volleyball research, but compared to sports like football or basketball, volleyball is still catching up. Most volleyball papers from the last two decades focus on classic statistical methods or manual expert analysis of various game situations, especially attack and serve (Afonso et al., 2012; Papadimitriou et al., 2004). These works showed that small changes in a team’s defensive formation or even the setter’s starting spot can impact whether the team successfully defends or not. However, these earlier studies didn’t really use machine learning as we know it today.

As technology improved and more video data became available, some researchers started experimenting with more modern approaches. For example, Marcelino et al. (2012) looked at factors like match status and opposition quality, combining several variables in models similar to what we now call machine learning, but without advanced algorithms. A real jump came with Zhao et al. (2017), who used deep learning to recognize volleyball actions in video footage. Their neural network could automatically identify spikes and blocks, showing that even with a limited dataset, supervised machine learning could outperform rule-based analyses.

Recently, research has moved closer to using machine learning for predicting outcomes and understanding tactics. Timi et al. (2022) built classifiers such as logistic regression and support vector machines to predict rally outcomes, not just recognize actions. They found that adding features about the positions and roles of players, and not just simple stats, helped the models perform better (up to 77% accuracy). This agrees with what Gudmundsson and Horton (2017) found in a broad sports data review: engineered spatial and context features-like distances between players or coverage of likely attack zones-matter much more than just raw numbers.

Another interesting angle is using unsupervised models to group typical formations. Tuma et al. (2023) applied k-means clustering on players’ positions and discovered that the resulting clusters often matched known volleyball formations, which can help coaches recognize team tendencies and weaknesses.

One big challenge in volleyball analytics is data. Unlike football, there aren’t big open datasets, so most research uses hand-annotated video or small samples. This makes careful feature engineering, clear labeling, and data validation very important. Many recent articles (Bouchlaghem et al., 2022) also warn about data leakage, where information from after the rally (like the exact landing zone) leaks into the predictions, making accuracy appear better than it is.

Despite these hurdles, it's clear that volleyball is starting to benefit from machine learning, especially in areas like action recognition, tactical evaluation, and outcome prediction. However, most systems are still only available to professional teams, so there is a big need for open datasets and tools that regular coaches and players can use. This thesis builds on these ideas by creating a custom dataset and focusing on features that describe the real tactical choices of defenders, with the goal of making ML-based analysis useful and practical at all levels of volleyball.

2. Theoretical Background

In this section, we briefly cover the basic principles of defensive formation in volleyball, focusing on how team positioning aims to cover the most likely attack zones and support successful transitions. Then, it introduces the fundamental concepts of machine learning relevant to this work.

2.1 Principles of Defensive Formation in Volleyball

A team’s defensive formation in volleyball plays a crucial role in determining the outcome of rallies when facing an opponent’s attack. The fundamental aim of any defensive system is to maximize court coverage, anticipate the direction of attacks, and position players so that the team has the best possible chance to dig powerful spikes or react to tips and roll shots. Defensive alignment is usually coordinated between the front row (blockers) and the back row (diggers), with each player assigned specific court zones and responsibilities. One specialized defensive player, the libero, can substitute freely in the back row to strengthen defense even more (WOVC, n.d.).

The two most common systems are the “6-back” defense, where the middle back-row defender covers deep attacks, and the “6-up” or “perimeter” defense, which brings the middle back-row defender closer to the net to guard against short balls and tips. The choice between these systems often depends on the opponent’s tendencies, the strengths of one’s own blockers and back-row players, and even the rotation—for example, when the libero is in a key back-row position (Papadimitriou et al., 2004). Successful defense depends not just on initial positioning, but also on players’ ability to quickly read the setter, react to the attacker’s movements, and continuously adjust spacing to avoid leaving open zones.

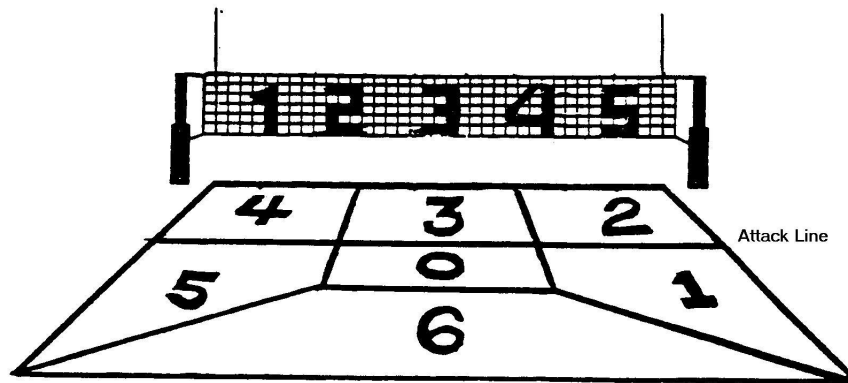


Figure 2.1: The Defensive Zone Grid (WOVC, n.d.)

2.2 Machine Learning Fundamentals

Machine learning (ML) is a branch of artificial intelligence focused on developing computational models that can learn from data and make predictions or decisions with minimal human intervention (Goodfellow et al., 2016). In recent years, machine learning has become an essential tool in sports analytics, including volleyball, where it is used to analyze player performance, predict match outcomes, and optimize tactical decisions (Timi et al., 2022).

2.2.1 Supervised Learning and Binary Classification

This thesis applies supervised learning, a paradigm in which algorithms are trained on labeled data pairs consisting of input variables (features) and corresponding target outcomes (labels) (Liu & Wu, 2012). The goal is for the algorithm to learn a mapping from inputs to outputs, so it can accurately predict outcomes for unseen data. In this thesis, the prediction target is binary: whether a specific defensive formation results in an "optimal" or "non-optimal" defensive outcome (binary classification).

Features are measurable variables that describe the situation before the opponent's attack, such as player positions (zones), set quality, attack type, block touch, and others. The label (target variable) indicates the actual outcome-success or failure of the defense.

2.2.2 Feature Engineering and Data Preparation

Careful feature selection and preparation are crucial for effective machine learning (Gudmundsson & Horton, 2017). In this thesis, only features that are available before the outcome are used, to avoid data leakage (i.e., ensuring the model cannot "cheat" by using future information). Categorical variables such as attack type are numerically encoded, and, for some algorithms, feature scaling is applied to standardize values.

Feature importance methods, especially in tree-based models, help to determine which input features most influence the model's decisions (Breiman, 2001). This not only boosts predictive performance but also provides tactical insight, which is especially valuable in sports analytics.

2.2.3 Machine Learning Algorithms used

Three main machine learning algorithms were implemented and compared for defensive outcome prediction:

- **Logistic regression:** A classic statistical model for binary classification, logistic regression estimates the probability that an input belongs to the positive class (e.g., optimal defense). It is based on the logistic (sigmoid) function, which outputs values between 0 and 1. Logistic regression is simple, interpretable, and serves as a strong baseline (Hosmer et al., 2013).
- **Random forest:** This is an ensemble method that builds multiple decision trees and combines their outputs. Each tree is trained on a random subset of the data and features. Random forest models can capture complex, nonlinear relationships and are less likely to overfit than single decision trees. They also provide feature importance scores, enabling interpretation of which features matter most (Breiman, 2001).
- **XGBoost classifier:** XGBoost is a powerful ensemble method based on gradient boosting. It constructs trees sequentially, with each new tree correcting errors made by the previous ensemble. XGBoost is recognized for its efficiency, regularization options (reducing overfitting), and strong performance in many applied research scenarios (Chen & Guestrin, 2016). Like random forest, XGBoost outputs feature importance metrics.

These three methods were chosen due to their proven effectiveness in sports analytics research and their suitability for small to medium-sized structured datasets (Timi et al., 2022).

2.2.4 Model Validation and Evaluation

A standard approach to evaluating model performance involves splitting the dataset into a training set and a test set. The model learns patterns from the training set and is then evaluated on the test set, which simulates new, unseen data. To further ensure robust evaluation, k-fold cross-validation is used: the training set is divided into k folds, and the model is trained and validated multiple times, each time using a different fold as the validation set (Berrar, 2018).

Hyperparameter tuning is performed using techniques like grid search to find the optimal model parameters (such as tree depth, number of estimators, learning rate, etc.) based on validation performance.

Model performance is assessed using several metrics:

- **Accuracy:** The proportion of all predictions that matched the true outcome.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (2.1)$$

- **Precision:** Of all predicted “optimal” cases, this is the share that were actually optimal.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2.2)$$

- **Recall:** Of all true “optimal” cases, the proportion identified by the model.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (2.3)$$

- **F1-score:** The harmonic mean of precision and recall, giving a balanced view.

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.4)$$

Additionally, confusion matrices were created to visualize the numbers of true positives, false positives, true negatives, and false negatives for each model. These evaluations allow a nuanced understanding of model strengths and weaknesses in the context of volleyball.

2.2.5 Probability Calibration

In binary classification, many models output a probability (e.g., “this formation has a 70% chance of optimal defense”). However, these probabilities are not always well-calibrated—that is, the predicted probability may not match the actual frequency of the event. To address this, calibration techniques such as Platt scaling (sigmoid) and isotonic regression are used (Niculescu-Mizil & Caruana, 2005). A calibration curve is plotted to visually assess how well predicted probabilities align with observed outcomes. Well-calibrated probabilities are especially important in sports analytics, as coaches may use these values directly for tactical decision-making.

3. Dataset description

3.1 Data Collection

Since there are currently no publicly available datasets with enough detail about player locations and game outcomes for volleyball, I decided to manually collect the data to make sure it would be both high-quality and relevant. For my analysis, I focused on professional matches, specifically using footage from the Paris Olympics 2024 game between Brazil and Poland, Paris Olympics 2024 game between France and Italy, Paris Olympics 2024 game between USA and Italy and Paris Olympics 2024 game between USA and Germany which I all found on YouTube. I only selected videos that were recorded with a fixed camera high above the court, so I could always see every player’s position clearly throughout each rally.

Each defensive situation was carefully reviewed frame by frame, and at the moment when the attacking player hit the ball at the net, I recorded the positions of all six defenders as well as the attacking player. To keep the annotation process consistent and systematic, I used a court grid (see in Fig. 3.1), where each cell has its own unique zone number. In this grid, the green squares represent the main playing area for both teams, while the blue zones indicate out-of-bounds regions and are useful for tracking when balls land outside the court. The solid red line marks the net and dotted red lines represent attack lines. This grid, adapted from the VREN study, makes it possible to compare player positioning and defensive coverage across different rallies in a reliable, data-driven way (VREN, 2023).

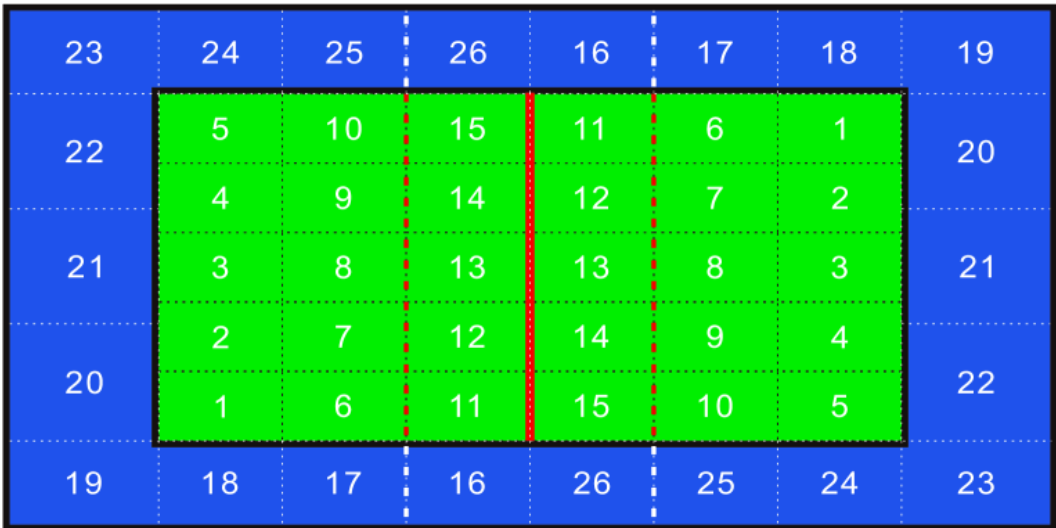


Figure 3.1: Grid layout of a volleyball court (VREN, 2023)

However, the process of manually annotating all aspects of every defense play turned out to be very time-consuming. In the beginning, the most crucial thing was to decide which aspects of the game have an impact on the success of the defense. During the process of creating the dataset I made some changes in feature selection and also in way how I was annotating the positions of players so I had to start all over again with creating the dataset. That's why I only managed to collect 300 samples. Still, I think that's enough for volleyball, since there isn't as much variety in strategy compared to some other sports.

3.2 Data Exploration

To better understand the structure of my dataset and ensure I was using the best variables for modeling, I first reviewed all available features (see in Table 3.1) and considered their relevance to the prediction of volleyball defense.

Since I was building the dataset manually, I encoded the categorical features directly as integers. This meant the dataset was already prepared for modeling and didn't require any extra pre-processing. I also chose not to include features like `attack_land_zone` in the predictive modeling, even though it provides a lot of useful information. The reason is that this feature reveals the outcome of the play, which would lead to data leakage.

Table 3.1: Description of all used attributes (Author's calculation)

Column	Value type	Description
<code>defense_player_1</code>	Integer	First defense player position
<code>defense_player_2</code>	Integer	Second defense player position
<code>defense_player_3</code>	Integer	Third defense player position
<code>defense_player_4</code>	Integer	Fourth defense player position
<code>defense_player_5</code>	Integer	Fifth defense player position
<code>defense_player_6</code>	Integer	Sixth defense player position
<code>set_quality</code>	Integer	Quality of the set
<code>attack_player_jump_zone</code>	Integer	Zone of attacking player
<code>attack_type</code>	Integer	Type of the attack
<code>block_touch</code>	Integer	Touch of the ball by blockers
<code>num_of_blocks</code>	Integer	Number of players blocking
<code>outcome</code>	Integer	Result of the defensive play

The main features I focused on are the positions of all six defenders, which show exactly where each player is just before the attack happens. I also included the starting zone of the attacker, since defensive formations often change depending on where the attack is coming from. For the attack type, I considered several options: `hit` (a powerful spike), `off_speed` (a slower spike), `roll_shot`, `tip`, `free_ball` (a passive return over the net), and `dump` (when the setter attacks). I added set quality as a feature because less accurate sets can make things easier for defenders. Finally, I recorded details about the block-specifically, how many blockers were involved and whether the block touched the ball-to see how the block affected defensive outcomes. Some of these features were adapted from the VREN study, as they were discussed with volleyball coaching experts and are considered reliable (VREN, 2023). To differentiate between zones on the attackers’ and defenders’ sides, I used a minus sign for zones on the attacking side. When annotating the positions of all defensive players, I started with the back row, marking players from right to left. I then did the same for the middle row, and finally, I annotated the positions in the front row.

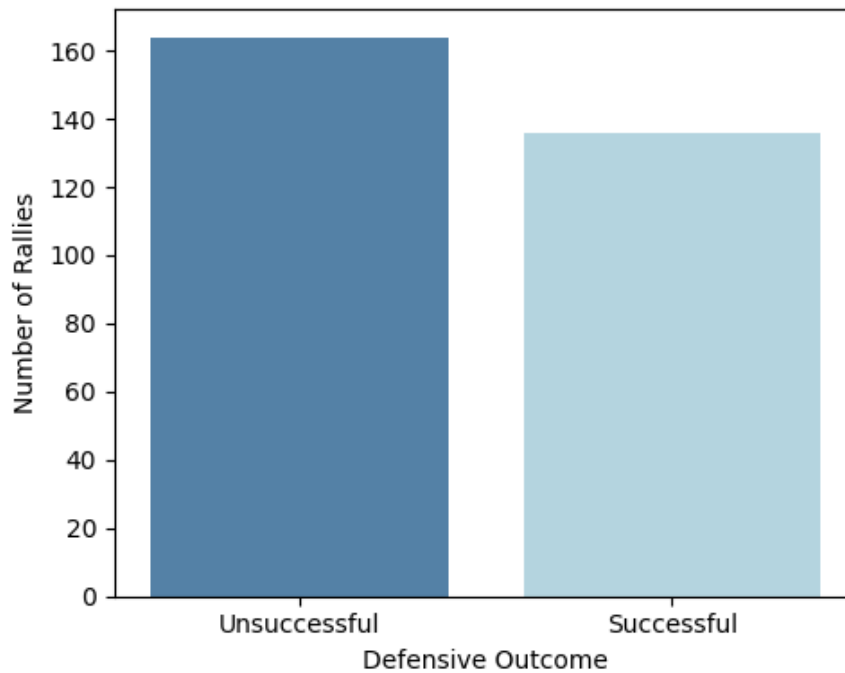


Figure 3.2: Distribution of Defense Outcome (Author’s calculation)

The bar chart above (see in Fig. 3.2) shows how often each defensive outcome occurred in my dataset. I split the rallies into two groups: “Successful” and “Unsuccessful” defenses. As the chart shows, there were more unsuccessful defensive plays (just over 160) than successful ones (about 135). From the chart, you can see that there were more unsuccessful defensive plays-just over 160-compared to about 135 successful ones. This gives a quick idea of the balance between the two results and shows how challenging it can be to pull off a successful defense in the professional volleyball matches I analyzed.

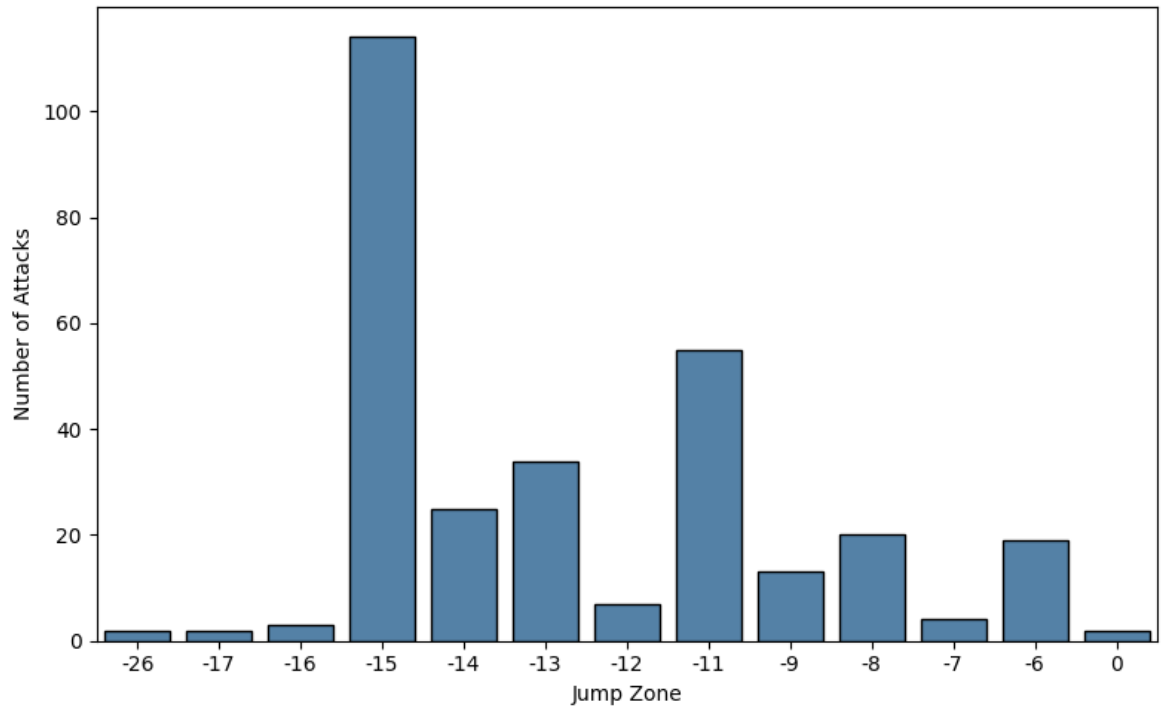


Figure 3.3: Distribution of Attacker Jump Zones (Author’s calculation)

As you can see in Figure 3.3, most attacks started from zone 15 (minus sign represents the side of the attackers), which is usually where the outside hitter attacks from. This zone had by far the highest number of jumps, over 100, which makes sense because a lot of sets go to the left front in high-level volleyball.

Other zones with a decent number of attacks are 13 and 11, which are also in the front row, specifically middle and left side. These are typical positions for quick attacks by the middle blocker or for the right-side hitter. There are also some attacks from zones further away from the net, like 6 or 8, but these are much less common, probably because back-row attacks are used less often.

4. Modeling

4.1 Choice of ML method

In this thesis, the goal is to predict whether a volleyball defensive formation will result in an optimal or non-optimal outcome (binary classification) based on information available at the moment of attack. For this purpose, I selected several supervised machine learning models that are well-suited for tabular data and binary outcomes:

- **Logistic regression**
- **Random forest**
- **XGBoost classifier**

These models were chosen specifically because they work well with relatively small datasets (300–500 samples), don't require deep learning or heavy computational resources, and offer insights into feature importance. Additionally, they are widely accepted in recent sports analytics research for binary classification problems (Timi et al., 2022). During preliminary experiments, all three models achieved fairly similar performance levels, likely due to the modest dataset size and the use of carefully engineered features. Ultimately, the XGBoost classifier achieved the highest accuracy and F1-score on the test set. Therefore, I selected XGBoost as the final model for more detailed evaluation and interpretation in this thesis. However, the difference in accuracy between the models was only a matter of a few correctly classified predictions.

4.2 Model training and testing

To train and compare the models, I first split the dataset into a training set (75%) and a test set (25%) using stratified random sampling, ensuring both optimal and non-optimal outcomes were proportionally represented in each subset. Categorical variables such as attack type were coded numerically, and for logistic regression, all input features were standardized using a scaler. I had to standardize input features for logistic regression, because logistic regression is sensitive to how features are scaled—if some variables have much larger numeric ranges than others, they can end up dominating the model even if they're not actually more important. By standardizing (so each feature has a mean of zero and a standard deviation of one), I made sure that all features contribute equally to the result. This helps the model learn better and makes the training process more stable. For tree-based models like Random Forest and XGBoost, scaling isn't necessary because those algorithms aren't affected by the scale of the features, so I only used the scaler for the logistic regression model.

Table 4.1: The final comparison of XGBoost model on the test set (Author’s calculation)

Accuracy	Precision	Recall	F1-score
0.75	0.76	0.65	0.70

For each model-logistic regression, random forest, and XGBoost-I carried out hyperparameter tuning using grid search with 5-fold cross-validation. This method systematically evaluated different combinations of key hyperparameters (such as tree depth, number of trees, and learning rate for XGBoost) to maximize the F1-score on the validation folds. After tuning, the best estimator for each algorithm was retrained on the full training set and then tested on the holdout test set.

Table 4.2: Parameter grid used for hyperparameter tuning of XGBoost, best hyperparameters are underlined (Author’s calculation)

Hyperparameter	Value
n_estimators	100, <u>200</u>
max_depth	<u>3</u> , 5, 7
learning_rate	<u>0.1</u> , 0.2
subsample	0.8, <u>1</u>

Despite testing a variety of parameter grids for XGBoost, I found that the best results were consistently achieved at the edge values of the grid. Although it is generally recommended to avoid having the optimal hyperparameters on the grid boundaries-since this can indicate the grid is too narrow or not well centered-expanding the grid in my case actually led to a decrease in accuracy. This likely reflects both the small size of my dataset and the sensitivity of XGBoost to specific parameter ranges when limited data is available. For small, structured datasets like mine, it is not uncommon for tree-based models to perform best with simpler or default settings, while more complex or extreme configurations can cause underfitting or instability. Therefore, for this project, I used the edge values that performed best, but I acknowledge that with larger datasets or different problems, a broader or more sophisticated search might be necessary to avoid this limitation (Probst et al., 2019; Timi et al., 2022).

4.3 Model Evaluation

4.3.1 Confusion Matrix

To better understand the performance of the selected XGBoost model, I analyzed its confusion matrices for test and training sets and feature importances. It is important to highlight that the test set represents 25% of the entire dataset.

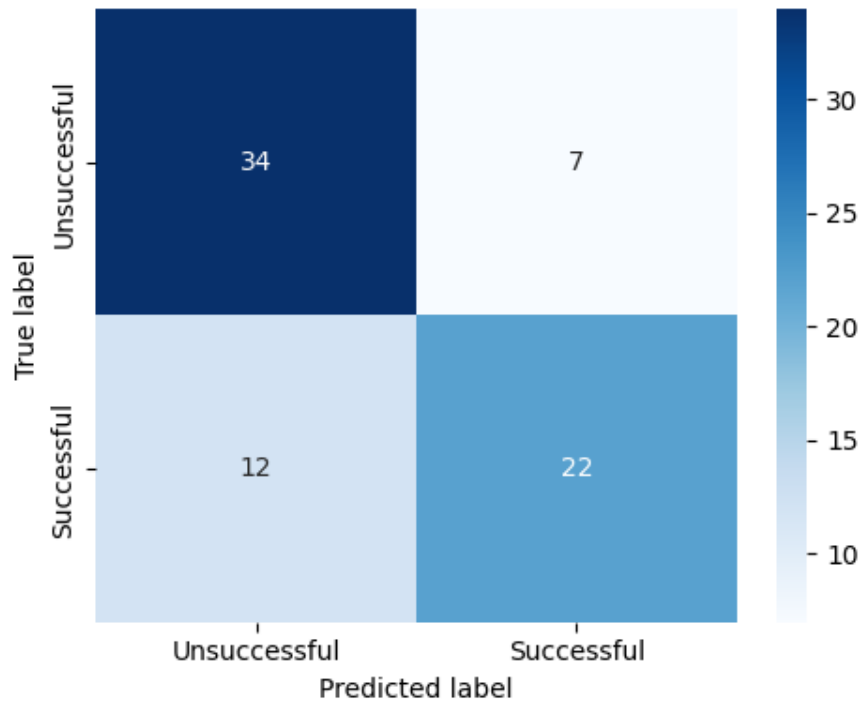


Figure 4.1: Confusion matrix for XGBoost (Author's calculation)

In the context of volleyball defense:

- **True Negatives (top left, 34 cases):** The model correctly identified unsuccessful defenses. This means it agreed with reality that the defending team failed to control the attack, either missing the dig or failing to block the ball.
- **False Positives (top right, 7 cases):** The model predicted successful defense, but in reality, the defense was not successful. Here, the model assumed the play would be defended, but the team did not manage to dig the ball or block the attack as expected—these are overestimates of the defense's ability.
- **False Negatives (bottom left, 12 cases):** The model predicted unsuccessful defense, but the team actually succeeded defensively. In these rallies, the model underestimated the defense—the team might have managed to dig a difficult attack or block the ball successfully, even though the model thought this was unlikely.

- **True Positives (bottom right, 22 cases):** The model correctly predicted that the defense would be successful. This reflects situations where the defending team either controlled the attack with a quality dig or blocked the ball, as anticipated by the model.

4.3.2 Model accuracy on training data

By comparing both matrices (Figure 4.1 and 4.2), it can be observed that the model performs slightly better on the training set (accuracy 0.83) than on the test set (accuracy 0.75). This gap indicates a mild degree of overfitting, which is common, but the difference is not extreme. More important is that the distribution of false positives and false negatives is similar across both sets, which shows the model’s predictions remain stable and reliable even on new data.

$$\text{Accuracy}_{\text{train}} = \frac{112 + 74}{112 + 11 + 28 + 74} = \frac{186}{225} \approx 0.83 \quad (4.1)$$

For coaches and players, this means that the model is more likely to correctly recognize unsuccessful defenses (true negatives) than to predict a successful defense (true positives), a property that is often valuable for minimizing unexpected defensive failures in game situations.

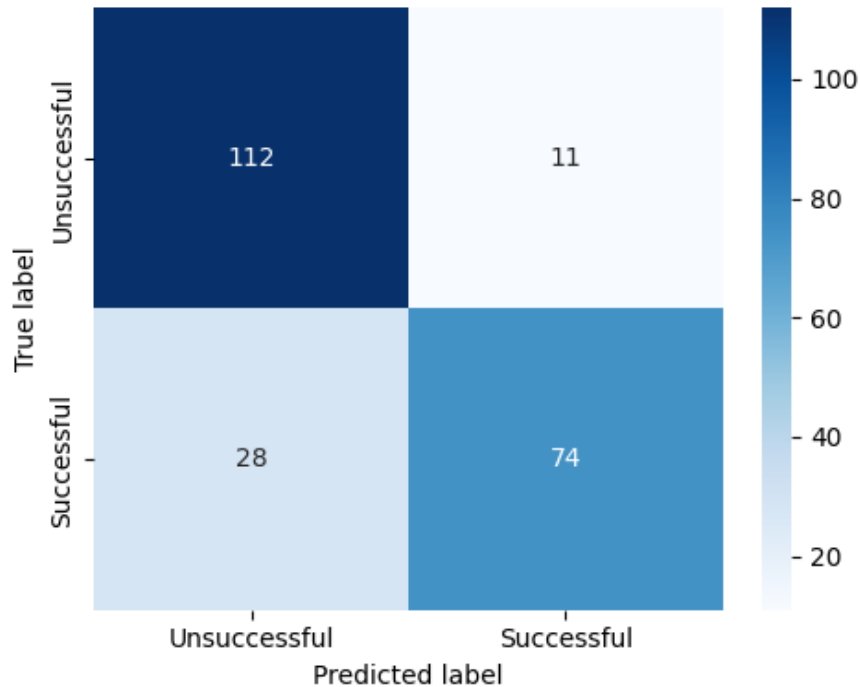


Figure 4.2: Confusion matrix for XGBoost - train set (Author’s calculation)

4.3.3 Feature importance

Feature importance analysis (see in Figure 4.3) reveals which variables most influenced the XGBoost model. The leading predictor was the type of attack (`attack_type`), which makes sense as different attacks (like spikes, tips, or roll shots) pose varying levels of difficulty for defenders. Block effectiveness (`block_touch` and `num_of_blocks`) and the position of the sixth defender (`defense_player_6`) were also key factors, showing that both net defense and certain player positions are crucial for a successful outcome. In contrast, some defender positions (such as defenders 2, 4, and 5) played a smaller role in the model's predictions.

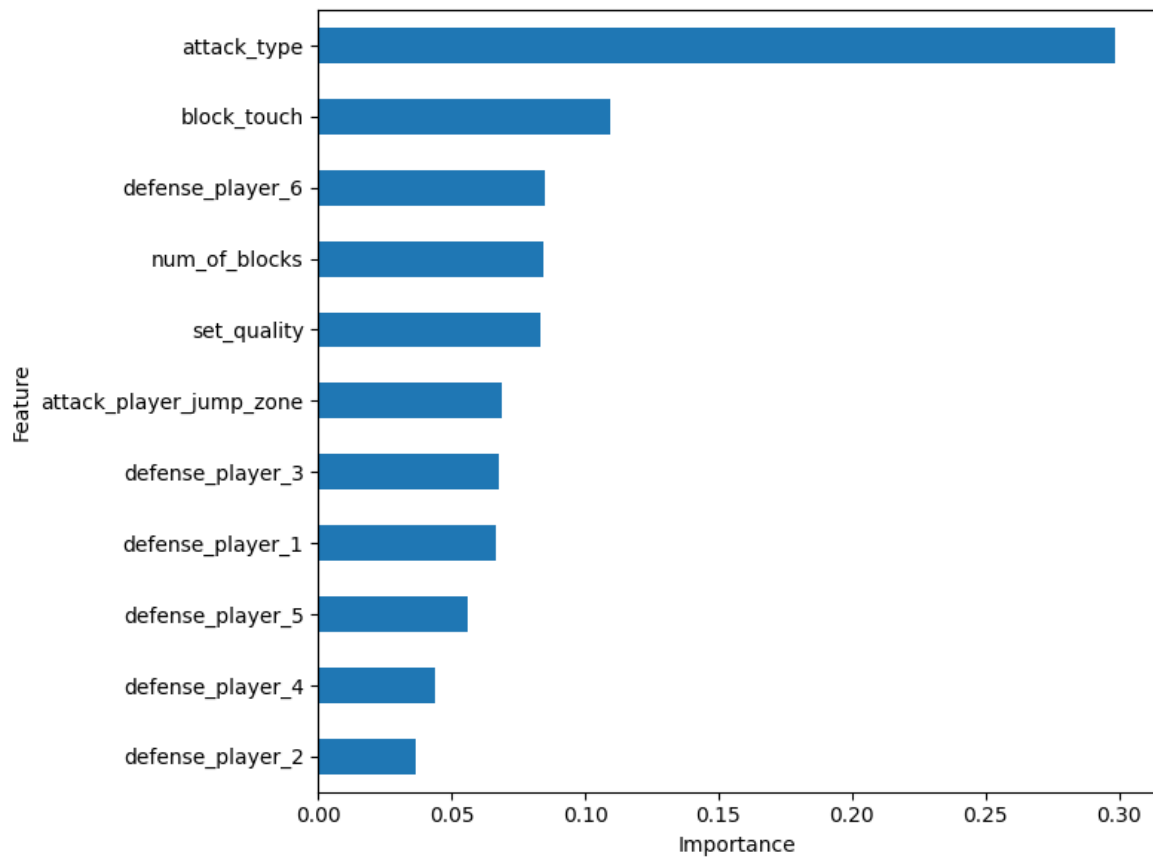


Figure 4.3: Feature importances for XGBoost (Author's calculation)

5. Analysis of defensive positions

5.1 Impact of Defense Player 6 position

To better understand how specific changes in team formation influence the probability of a successful volleyball defense, I decided to use a logistic regression model. It is especially suitable for my application because it not only predicts whether a defense is likely to be optimal or not, but also directly estimates the probability of a successful outcome based on the input features. This probability can be interpreted as “the chance of an optimal defense,” which is exactly the kind of actionable information coaches can use. I specifically chose to analyze the impact of defense player 6 because, according to my feature importance analysis, this defender’s position had the greatest importance among all player positions in predicting defensive success. This suggests that where player 6 stands may have a significant effect on the team’s overall defensive effectiveness, making it a particularly interesting variable to investigate in detail.

5.1.1 Getting Probabilities from logistic regression

Once the logistic regression model is trained on the data, it outputs a probability for each prediction. This is done using the sigmoid function, which transforms the model’s linear combination of features into a value between 0 and 1. In Python, this is accessed through the `predict_proba` method, where the second column gives the probability of the positive class (in my case, optimal defense). For example, if the model returns 0.7, it predicts a 70% chance that the given defensive setup will succeed.

5.1.2 Calibration Process

However, it is important that these predicted probabilities actually match the real-world chances. In practice, the raw outputs from logistic regression are not always perfectly aligned with actual outcome frequencies, especially with limited or imbalanced data. To check this, I plotted a calibration curve (see in Figure 5.1, blue). Ideally, if the model predicts a probability of 0.6, then out of all those situations, 60% should actually be successful defenses.

In my initial calibration curve, I could see that the model’s probabilities did not always track the real-world frequencies closely. Some points fell above or below the diagonal, indicating overconfidence or underconfidence at certain probability levels. To address this, I used isotonic regression calibration (Berrar, 2018), which adjusts the predicted probabilities so that they better reflect observed outcomes.

After calibration, I plotted a new calibration curve (see in Figure 5.1, orange). Now, the curve followed the diagonal much more closely, showing that the adjusted probabilities were much more reliable. This step was crucial: it means my model's 70% prediction really does represent a roughly 70% chance of defensive success, making the analysis both trustworthy and interpretable.

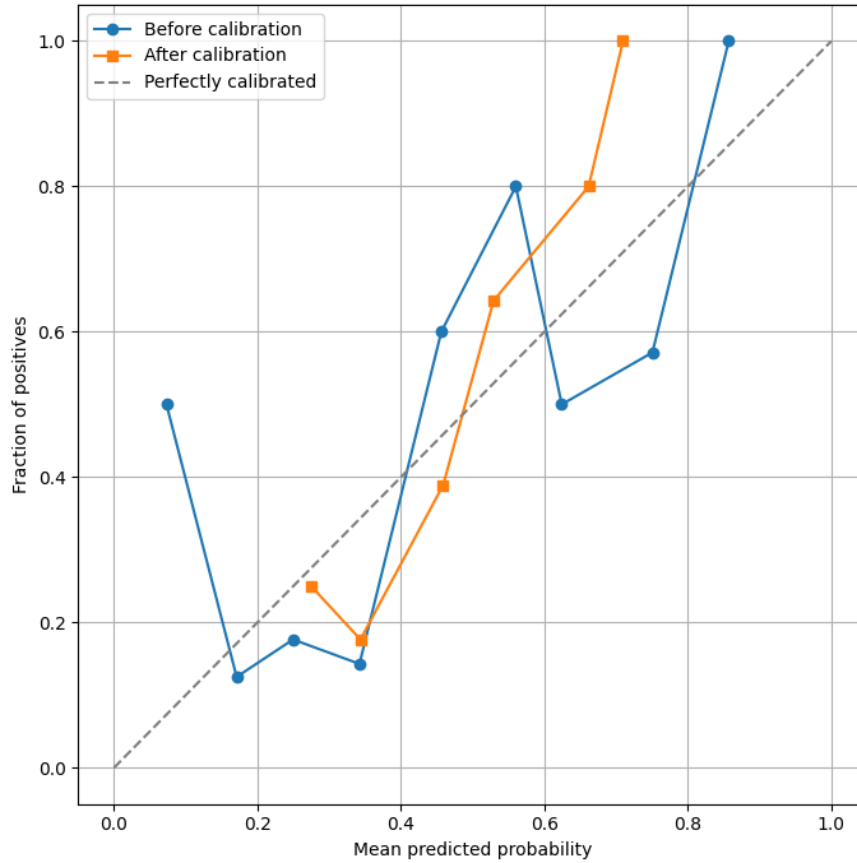


Figure 5.1: Calibration Curves - Before and After Calibration (Author's calculation)

5.1.3 Results

With a well-calibrated model, I proceeded to analyze how the position of defender 6 affects the probability of successful defense. I created a typical rally situation based on the most common values from my dataset and then systematically varied only the zone of `defense_player_6`, keeping all other features fixed. For each position, I used the calibrated model to obtain the probability of success.

The results, shown in Figure 5.2, reveal a clear pattern: the probability of successful defense is highest when defender 6 is placed in zones 11 or 12 (right or right-center front row), and decreases as this player moves toward the left side of the net. For example, the predicted probability drops from about 0.33 in zone 11 to 0.27 in zone 14.

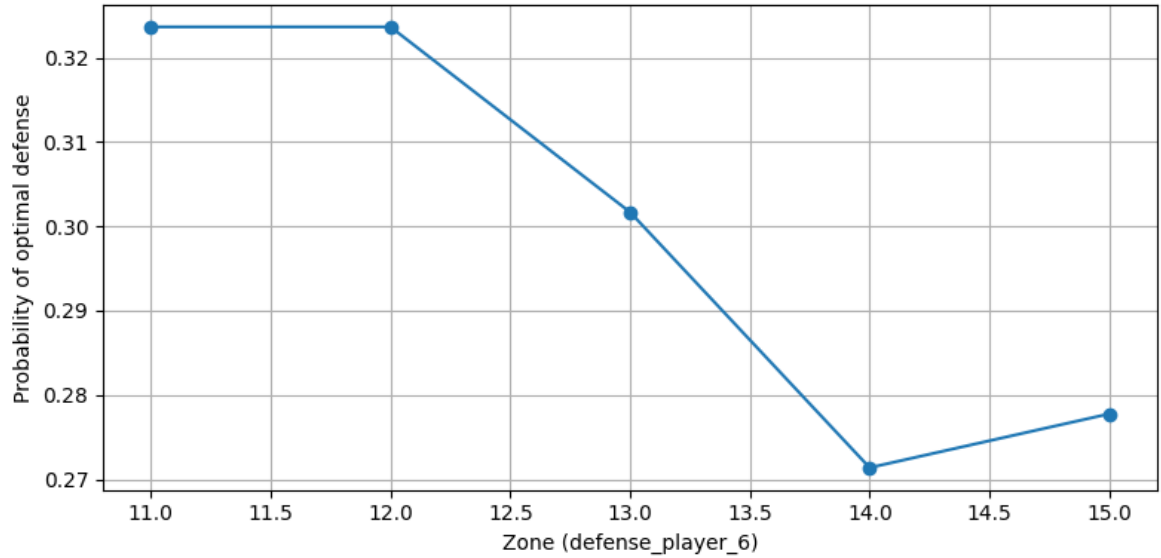


Figure 5.2: Probability of Successful Defense by Defender 6's Zone without Block touch (Author's calculation)

On the other hand, when I set the 'block_touch' feature to 1-which means the blockers actually touched the ball-the pattern was quite different. As you can see in Figure 5.3, the probability of a successful defense actually goes up as defender 6 is moved from the right front (zone 11) towards the left front (zone 15). It starts at about 0.42 and increases to around 0.47. This means that when the block is working and manages to slow down or deflect the attack, having defender 6 further to the left really helps. Most likely, this gives the team a better chance to react to the changed trajectory of the ball, cover tips or quick kills, and pick up any rebounds off the block. So, in these situations, it seems like putting defender 6 on the left front side is a good tactical move.

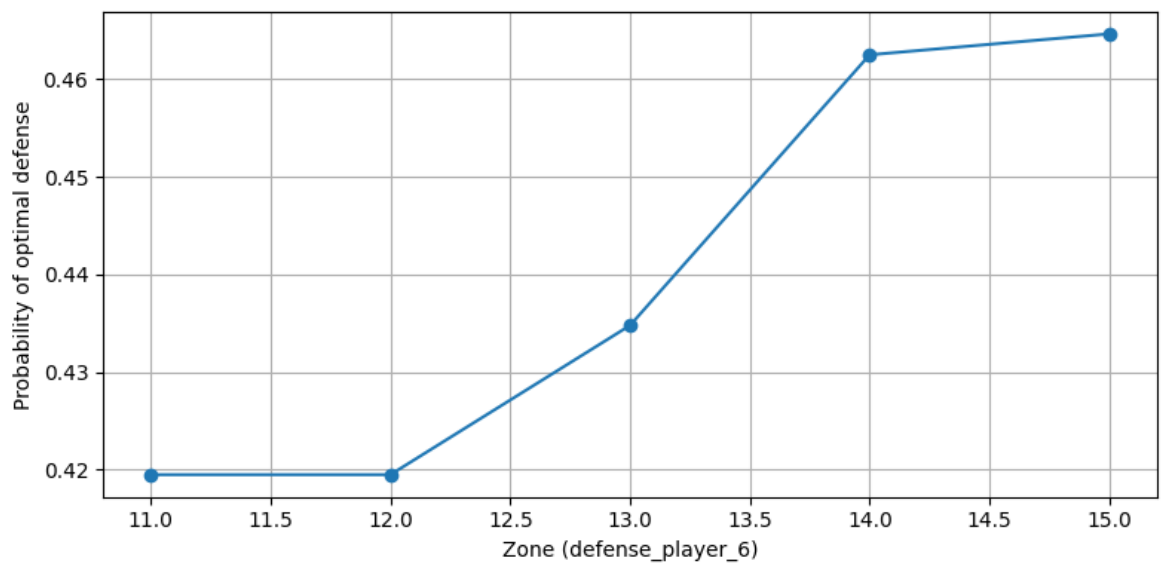


Figure 5.3: Probability of Successful Defense by Defender 6's Zone with Block touch (Author's calculation)

These differences reflect typical volleyball defensive dynamics. A block touch significantly alters the play, often forcing attackers to adjust or making the ball easier for defenders to control behind the net. Hence, when blocks are successful in contacting the ball, defender 6's movement toward the left front zones enhances team defense, possibly by covering tips, quick attacks, or rebounds. In contrast, when no block touch occurs, the defense faces a stronger direct attack, requiring player 6 to position more conservatively or optimally on the right front side, where the majority of attacks might be directed or where coverage is traditionally stronger for that team.

5.2 Heatmap Analysis of Defensive Formations

Understanding how defenders position themselves in response to different attack patterns is crucial for effective volleyball tactics. To explore these patterns, I created heatmaps showing the most frequent zones occupied by defenders, focusing on the three most common attack jump zones in my dataset (see in Figures 5.4, 5.5, 5.6). Each heatmap represents a snapshot of the defending team's spatial tendencies during specific types of attacks.

5.2.1 Attacks from Zone 15



Figure 5.4: Defender Heatmap for Attack Jump Zone 15. The large number shows the number of occurrences; the small green number indicates the zone label (Author's calculation)

In the first heatmap (Figure 5.4), which corresponds to attacks initiated from jump zone 15, we see extremely strong clustering in the back-left (zone 11: 165 occurrences) and middle-back (zone 12: 92) of the court. The front-middle (zone 3: 67), mid-right (zone 9: 98), and mid-left (zone 6: 97) also show significant occupation. This suggests that when the attack originates from this zone, teams tend to reinforce the left-back and central defensive positions, likely anticipating cross-court hits or sharp angles. The clear emphasis on this side, with high counts along the left-hand zones, aligns with standard volleyball tactics where defenses shift to address strong left-side attacks.

5.2.2 Attacks from Zone 11

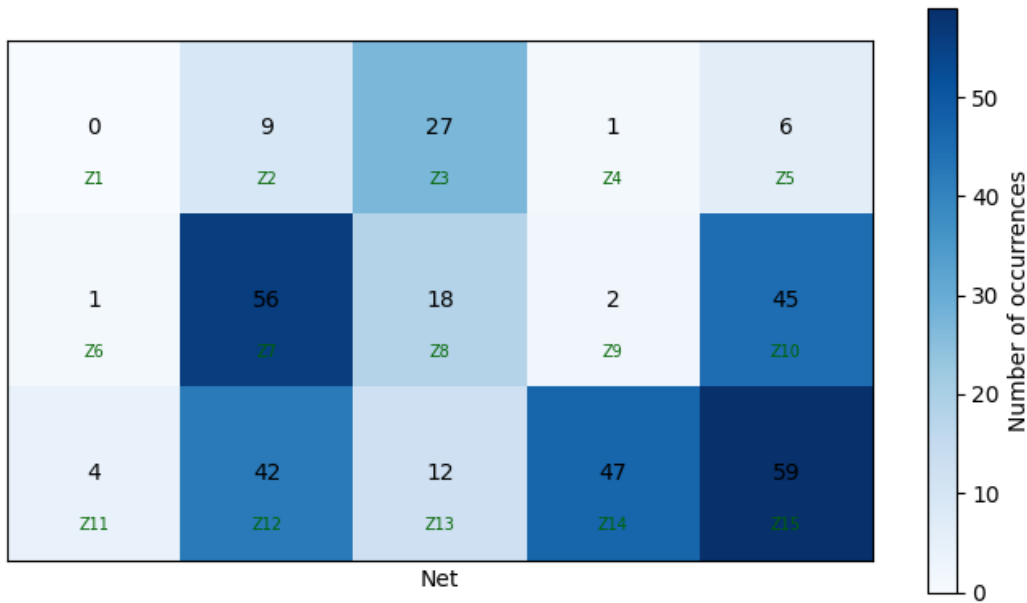


Figure 5.5: Defender Heatmap for Attack Jump Zone 11. The large number shows the number of occurrences; the small green number indicates the zone label (Author’s calculation)

Figure 5.5 focuses on the second most frequent attack jump zone (11). Here, the spread of defender positions is noticeably more balanced. While the left-back (zone 15: 59), right-back (zone 11: 42), and left-middle (zone 7: 56) attract higher concentrations, there is also a significant presence in the front-middle (zone 3: 27) and mid-right/back-right (zones 10 and 14). This distribution indicates that when the attack comes from 11, the defense adapts to cover a broader range of possible attack directions, reflecting a need for more balanced coverage against a potentially unpredictable attack trajectory.

5.2.3 Attacks from Zone 13

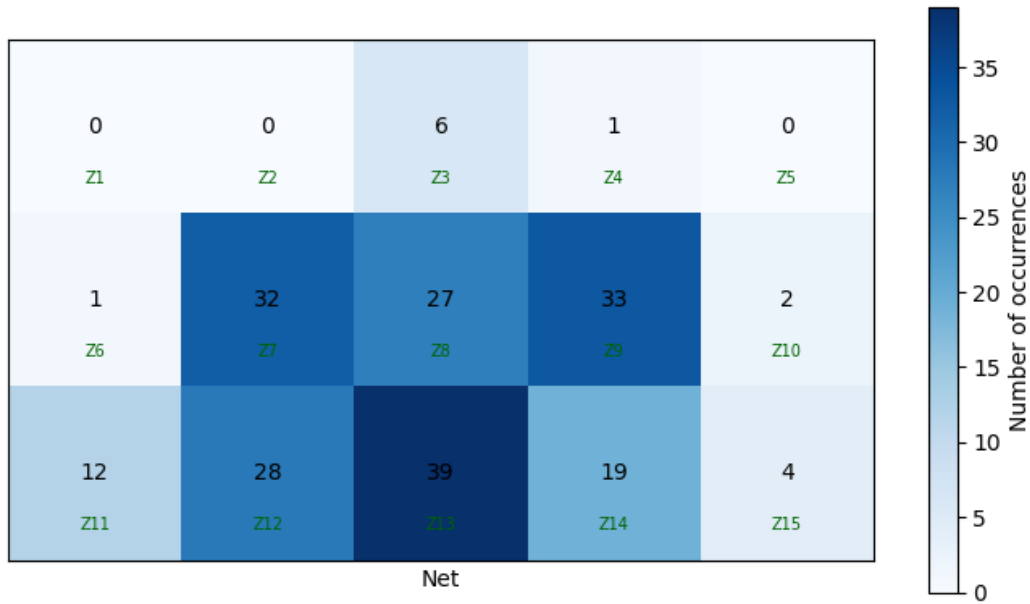


Figure 5.6: Defender Heatmap for Attack Jump Zone 13. The large number shows the number of occurrences; the small green number indicates the zone label (Author's calculation)

The third heatmap (Figure 5.6), for attacks from zone 13, demonstrates a high density in the very center (zone 13: 39) and adjacent left and right middle areas (zones 7, 8, 9, 12, 14). This suggests that, against attacks starting from 13, defenders prioritize the center and both flanks in the back line, possibly to counter quick offenses or pipe attacks that target gaps in the middle of the court.

Conclusion

The findings of this thesis demonstrate that machine learning models can provide valuable insights into the tactical effectiveness of volleyball defensive formations, even when trained on a relatively small and manually annotated dataset. Among the three models evaluated—logistic regression, random forest, and XGBoost—XGBoost achieved the highest accuracy (0.75) and F1-score (0.70) on the test set. However, the differences in performance between the models were not substantial, suggesting that for structured sports data of limited size, simpler models such as logistic regression can be nearly as effective as more complex ensemble methods. This is an important consideration for practical applications, as simpler models are often easier to interpret and explain to coaches and players.

A key aspect of this research was the careful selection and engineering of features. By excluding variables that could leak information about the outcome (such as post-attack events) and focusing solely on attributes available before the attack—like defender positions, attack type, set quality, and block details—the models were able to make predictions that reflect real-world tactical decisions. The analysis of feature importance confirmed that the type of attack, block effectiveness (`block_touch` and `num_of_blocks`), and especially the position of defender 6, were the most influential factors in predicting defensive success. This finding aligns with established volleyball coaching strategies, which emphasize the significance of effective blocking and the critical role of the middle back defender.

The calibration of probability outputs, particularly for the logistic regression model, proved to be essential for practical use. Initially, the model's probability estimates were not well-aligned with actual outcomes, which could lead to misleading tactical decisions. By applying isotonic regression for probability calibration, the model's predictions became much more reliable, ensuring that a predicted probability (e.g., 70%) closely matched the observed success rate. This step is crucial if such models are to be used in real-time decision-making by coaches and analysts.

The detailed analysis of defender 6's positioning provided actionable tactical recommendations. When the block did not touch the ball, positioning defender 6 in the right or right-center zones (zones 11 or 12) was associated with the highest defensive success, likely because these areas are most vulnerable to strong, direct attacks. In contrast, when the block was effective, shifting defender 6 toward the left front (zone 15) increased the probability of a successful defense, possibly due to better coverage of tips, deflections, and off-speed shots. These results highlight the need for dynamic and context-dependent defensive strategies, rather than static formations.

Heatmap analyses further illustrated how defensive alignments shift in response to different attacking zones. For attacks originating from the left front (zone 15), defenders tended to cluster in the left and center back positions, anticipating cross-court spikes. Attacks from the middle (zone 13) led to more central defensive coverage, likely to counteract quick attacks and pipe plays. These patterns confirm that professional teams already employ adaptive defensive strategies, and that machine learning can help quantify and visualize these tendencies, providing valuable feedback for tactical planning.

Despite these promising results, several limitations must be acknowledged. The dataset was relatively small (300 samples) and limited to a few professional matches, which may restrict the generalizability of the findings and the model's ability to capture rare or highly complex tactical situations. The manual annotation process, while precise, was time-consuming and could introduce subjective bias. Future research should focus on automating data collection-potentially using computer vision techniques-expanding the dataset to include more teams and competitions, and incorporating additional features such as player movement trajectories or contextual factors like score and rotation.

In summary, this thesis demonstrates the potential of machine learning as a tool for tactical analysis in volleyball. While there are limitations to the current approach, the results suggest that data-driven methods can complement traditional coaching strategies and provide new perspectives on defensive effectiveness. With further development and larger datasets, these techniques could play an increasingly important role in real-time tactical support and performance optimization in volleyball.

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